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From design to co-nature'ing: toward a metabolic epistemology of design in the age of human–nature–intelligence ecologies

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Abstract

Contemporary design operates within conditions of ecological destabilization and distributed technological agency that challenge anthropocentric and instrumental paradigms. While participatory, regenerative and AI-mediated design approaches have expanded disciplinary scope, their integration remains conceptually fragmented. This paper introduces Co-nature'ing as a positioning framework that reconceptualizes design as a metabolic epistemology – a relational mode of knowing and acting across Human–Nature–Intelligence ecologies. Grounded in process philosophy, complexity theory, ecological cognition and autopoietic systems theory, the framework advances a symmetrical triad in which biological systems and algorithmic infrastructures function as co-agents rather than resources or tools. The Co-nature'ing spiral – Co-Sensing, Co-Growing, Co-Evolving and Co-Flourishing – is presented as an operative translation of this epistemological shift. A pedagogical instantiation through the BIP Design Roadmap illustrates how distributed agency can be enacted within studio practice. The paper contributes a unified theoretical foundation for biodesign as cognitive ecology and relational co-evolution.

Impact statement

This paper introduces **Co-nature'ing** as a conceptual and pedagogical framework that reconceptualizes design practice as a relational process within Human–Nature–Intelligence ecologies. By integrating ecological thinking, living materials experimentation and speculative design approaches, the framework proposes a metabolic epistemology in which biological systems, technological infrastructures and human practices are understood as co-evolving agents within design processes. The study demonstrates the pedagogical potential of this approach through the implementation of a Blended Intensive Programme (BIP), where biomaterial experimentation with algae and speculative biofutures activities enabled students to explore alternative relationships between design, living systems and emerging technologies. The findings suggest that Co-nature'ing can contribute to the development of biodesign education by fostering ecological literacy, future-oriented thinking and relational design practices capable of engaging with complex socio-ecological challenges.

Introduction - design in a condition of planetary interdependence

The historical trajectory of design is inseparable from the epistemological architecture of modernity. Emerging as an applied form of rational organization, design was shaped by the logic of projection, optimization and human-centered intention (Cross 2006; Simon 1969). Its operative structure presupposed a separation between subject and object, designer and user, culture and nature. Within this framework, the world appeared as material to be shaped, systems to be optimized and futures to be engineered.

Yet the conditions within which design now operates fundamentally challenge this separation. Climate instability, biodiversity collapse, algorithmic mediation and socio-technical acceleration reveal a world characterized less by stability than by dynamic interdependence. These crises are not isolated failures but manifestations of what Morin (1990) described as complex systemic entanglement, a condition in which phenomena cannot be understood independently of the networks that sustain them. Similarly, Prigogine and Stengers (1984) work on dissipative structures demonstrates that living systems persist not through equilibrium, but through continuous adaptation within far-from-equilibrium conditions.

Such insights destabilize the classical image of design as external intervention. If reality is processual rather than static, relational rather than discrete, then design cannot remain an act imposed upon the world from outside its flows. It must be reconceived as participation within them.

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Process philosophy, as articulated by Whitehead (1929), offers a foundational lens for this repositioning. In Whitehead's account, entities are not substances but events constituted by relations. Existence unfolds as becoming. Design, viewed from this perspective, ceases to be the arrangement of inert materials and becomes the modulation of relational processes. Similarly,

Margulis's (1998) theory of symbiogenesis reframes evolution as cooperative integration rather than purely competitive selection, suggesting that life advances through interdependence. The implications for design are profound: innovation is not domination over systems, but integration within them.

Parallel developments in epistemology reinforce this relational turn. Peirce's theory of abduction foregrounds hypothesis formation as an iterative, interpretive act emerging from engagement with phenomena rather than detached deduction. Schön (1983) extended this insight into design practice through the concept of reflection-in-action, emphasizing that knowing arises within situated experimentation. Bateson (1972) further complicates the picture by describing mind itself as ecological, distributed across circuits of interaction rather than confined to the individual subject. Knowledge, in this sense, is systemic pattern recognition within relational fields.

When combined with the concept of autopoiesis proposed by Maturana and Varela (1980), which defines living systems as self-producing networks structurally coupled with their environments, the ontological and epistemological consequences become evident: agency is distributed, cognition is relational and adaptation is co-constitutive.

Design theory has partially absorbed these insights. Participatory design redistributed authorship beyond the professional designer (Bødker 2018). Regenerative design reframed responsibility toward ecological vitality. Intelligent design introduced algorithmic systems into creative processes, expanding cognitive mediation. Yet these developments often remain compartmentalized – social participation, ecological awareness and technological intelligence treated as adjacent but not structurally integrated domains.

The present condition requires more than parallel paradigms. It demands an epistemological synthesis.

Recent design scholarship has increasingly challenged anthropocentric assumptions by advancing more-than-human, posthuman and relational design frameworks. Giaccardi and Redström (2020) argue that technologies and artifacts must be understood as co-performers rather than passive instruments, destabilizing the primacy of human agency. Forlano (2017) similarly situates design within posthumanist discourse, emphasizing entanglement between bodies, infrastructures and environments. In the context of biomaterial research, Karana et al. (2023) propose living artifacts as active participants in regenerative ecologies, while Nilsson et al. (2024) explore human–plant interaction as distributed co-creation across species boundaries. Parallel discussions in AI-mediated design question the stability of authorship and foreground hybrid intelligence systems (Bown 2021; Verganti et al. 2020). Co-nature'ing situates itself within this expanding discourse but proposes a synthetic integration: rather than treating ecological agency, participation and intelligence as adjacent debates, it frames them as structurally interdependent dimensions of a unified epistemology.

However, the integration of computational systems within ecological design also introduces the risk of technocratic drift, in which optimization logics displace ecological attentiveness, embodied judgment and relational responsibility. Co-nature'ing therefore positions computational mediation within ecological accountability rather than technological acceleration alone.

The question guiding this paper is therefore not how design can become more sustainable, participatory or intelligent in isolation, but how it can be reconceptualized as a unified ecology of co-evolving agencies.

Co-nature'ing emerges within this horizon as a conceptual proposition. It extends participatory logics beyond the human sphere and positions nature and intelligence not as contexts or instruments but as co-agents. Rather than treating biological systems as resources and computational systems as tools, Co-nature'ing proposes a symmetrical relational structure – Human, Nature, Intelligence – in which design functions as mediator of exchanges, translations and adaptive flows.

The term Co-nature'ing intentionally foregrounds nature not as a privileged category but as a historical absence within modern design epistemologies. While human intentionality and technological intelligence have long occupied central positions within design discourse, ecological systems have frequently remained instrumentalized as passive context or resource. Co-nature'ing therefore does not subordinate Human or Intelligence dimensions to Nature, but restores ecological agency as a co-constitutive participant within design cognition.

In this framework, design becomes metabolic: it circulates meaning, matter and information across domains. It is neither purely social practice nor technical method, but a relational infrastructure of learning.

The Co-nature'ing spiral, articulated as Co-Sensing, Co-Growing, Co-Evolving and Co-Flourishing, translates this epistemological stance into operative form. Each phase represents not a linear step but a mode of relational engagement: sensing as distributed perception, growing as collaborative experimentation, evolving as adaptive integration and flourishing as regenerative stabilization.

The argument developed in this paper situates Co-nature'ing within a broader philosophical lineage while positioning it as a contemporary response to planetary interdependence. A bounded pedagogical instantiation through the BIP Design Roadmap will be briefly examined to demonstrate how such an epistemology can be enacted within design education.

If modern design was grounded in projection and control, the present moment calls for modulation and co-evolution. Co-nature'ing does not seek to replace existing paradigms but to weave them into a relational fabric adequate to the ecological and cognitive complexity of the twenty-first century.

Design, under this view, becomes less an act of authorship and more an act of mediation, an ongoing negotiation between human intention, biological vitality and machinic cognition.

Theoretical foundations: toward a metabolic epistemology

Co-nature'ing emerges from the recognition that design can no longer operate within an ontology of separation. The classical model of design – in which a human subject intentionally shapes passive material toward predefined ends – presupposes stable boundaries between culture and nature, cognition and matter, subject and object. In conditions defined by ecological instability and distributed technological agency, these separations become increasingly untenable. Biodesign, in particular, confronts designers with living systems, evolving materials and hybrid intelligence infrastructures that resist reduction to instrumental frameworks.

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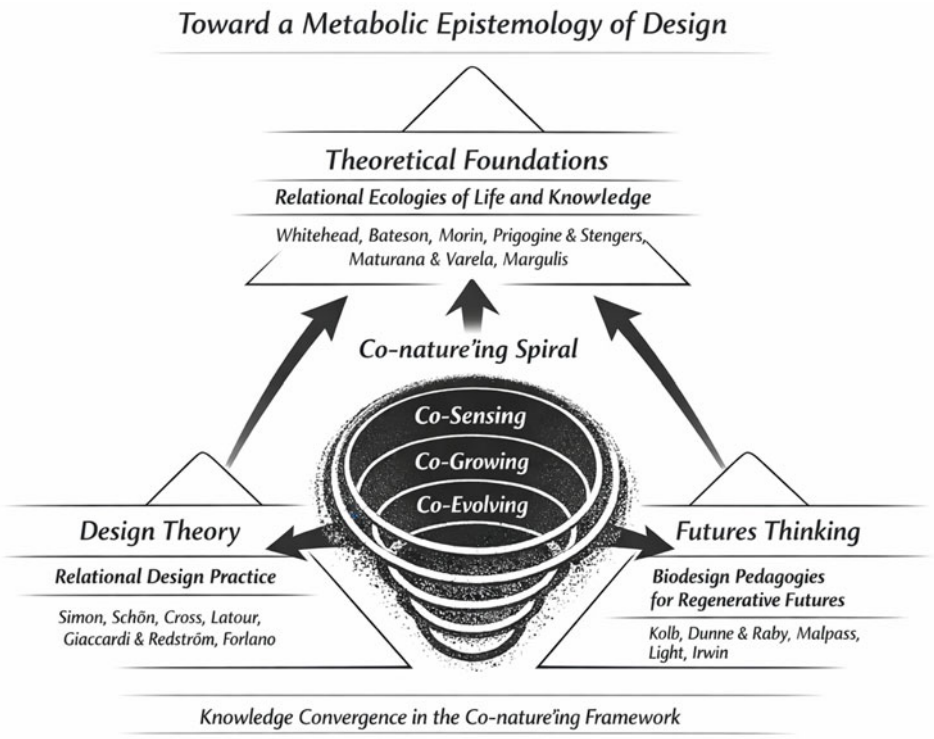


Figure 1. Toward a metabolic Epistemology of design, developed by Américo Mateus and Susana Leonor.
Source: The Authors

Process philosophy provides a foundational lens for rethinking this condition. Whitehead's (1929) conception of reality as relational becoming rather than fixed substance destabilizes the notion of discrete entities awaiting design intervention. In a processual ontology, entities exist as events within dynamic relational fields. Design, therefore, cannot be framed as the imposition of order upon inert substrates; it participates in ongoing transformations. This shift from substance to process reframes the artifact not as finished object but as temporary stabilization within broader flows of matter and meaning.

Complexity theory reinforces this ontological repositioning. Prigogine's work on far-from-equilibrium systems demonstrates that order emerges through dynamic instability rather than static equilibrium. Living systems persist through feedback loops, adaptation and fluctuation. When biodesign engages biological organisms or biomaterials, it operates within such dissipative conditions. Stability becomes provisional, negotiated through continuous recalibration rather than secured through deterministic control.

Evolutionary biology extends this relational logic. Margulis' theory of symbiogenesis challenges competitive models of evolution by demonstrating that new forms emerge through cooperative integration of previously distinct systems. Innovation, from this perspective, is not the result of isolated creativity but relational incorporation. Applied to design, this suggests that creative outcomes arise through integration across heterogeneous agencies – human intention, biological vitality and technological mediation – rather than from singular authorship.

These ontological considerations necessitate epistemological revision. Peirce's (1931–1958) concept of abduction positions knowledge as hypothesis formation arising from surprising encounters rather than deduction from fixed principles. Design activity, especially in biomaterial experimentation, exemplifies such abductive reasoning. Propositions are generated in response to material feedback and ecological constraint. Schön's reflective practice further articulates this iterative engagement, emphasizing that design knowledge is produced within action rather than prior to it. The designer does not merely apply knowledge but generates it through situated interaction.

Bateson's notion of an ecology of mind expands cognition beyond individual consciousness, locating intelligence within circuits of relational difference. This distributed model of cognition aligns with contemporary understandings of socio-technical systems in which human perception, material affordances and algorithmic processes co-constitute outcomes. Maturana and Varela's concept of autopoiesis reinforces this insight by describing living systems as self-producing networks structurally coupled with their environments. Structural coupling implies mutual adaptation: organism and environment co-determine one another's trajectories. Within such a framework, design becomes a coupling mechanism – mediating interactions among multiple co-evolving systems (Figure 1).

While these philosophical foundations establish relational ontology, Co-nature'ing also positions itself within contemporary design discourse. Participatory design redistributed authorship across human stakeholders, foregrounding collaborative processes. However, it frequently retains anthropocentric boundaries, limiting agency to social actors. Regenerative design shifted focus toward ecological restoration and systemic resilience, yet often maintains a teleological orientation in which human actors remain primary decision-makers. Posthuman and more-than-human design scholarship challenges human exceptionalism and foregrounds non-human participation (Giaccardi and Redström 2020; Karana et al. 2023). Nevertheless, these approaches sometimes lack a unified operative structure capable of integrating ecological agency with technological mediation and pedagogical translation.

Similarly, AI-mediated design research introduces distributed creativity and hybrid cognition (Bown 2021; Verganti et al. 2020).

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While these developments expand the scope of agency, they risk privileging computational generation without adequately situating it within ecological and ethical contexts. Co-nature'ing differentiates itself by structurally integrating these streams rather than treating them as parallel evolutions. It does not simply add ecological awareness to participatory design, nor append AI to regenerative practice. Instead, it proposes an epistemological synthesis.

While the Co-nature'ing spiral presents a terminological resonance with Scharmer's Theory U (2009), the framework departs from it in both ontological grounding and operative orientation. Theory U primarily addresses organizational transformation through human-centered social processes, whereas Co-nature'ing emerges from process philosophy, autopoiesis, ecological cognition, biomaterial experimentation and more-than-human design discourse. Its central concern is not organizational alignment but the relational integration of Human, Nature and Intelligence within evolving ecological systems. The recurrence of "Co-" terminology should therefore be understood not as direct methodological inheritance, but as part of a broader relational turn across contemporary systems-oriented design and transformation practices.

This synthesis is articulated through the Human–Nature–Intelligence triad. Unlike frameworks that prioritize one domain – social inclusion, ecological restoration or technological augmentation – the triad operates as a distributed but asymmetrical relational structure. Human intentionality, biological vitality and algorithmic mediation participate through different capacities, temporalities and forms of influence. Human actors retain ethical accountability and reflective mediation; biological systems participate through ecological and material agency; and computational systems contribute predictive, generative and analytical capacities that reshape perception and decision-making. Human intentionality, biological vitality and algorithmic mediation are understood as interdependent dimensions of distributed cognition. Design functions as mediator among them.

The term "metabolic epistemology" encapsulates this position. Metabolism implies continuous transformation and exchange. Just as biological organisms metabolize energy and matter, design metabolizes meaning, material and information across domains. Knowledge is not static representation but dynamic circulation. Decisions reverberate across ecological systems; material experimentation generates new epistemic conditions; computational modeling reshapes perception (Hui 2020).

Importantly, this metabolic framing does not dissolve ethical responsibility. On the contrary, it intensifies it. Leopold's land ethic extends moral consideration to ecological systems, suggesting that integrity and stability of the biotic community become evaluative criteria. Gilligan's ethics of care foreground relational responsibility rather than abstract universality. Co-nature'ing incorporates these insights without prescribing normative uniformity. Instead, it situates evaluation within relational consequences rather than isolated performance metrics.

In this way, Co-nature'ing addresses a gap in contemporary biodesign discourse. While existing frameworks highlight ecological integration or distributed intelligence, few provide a unified epistemological grounding capable of coherently integrating living materials, social collaboration and AI-mediated design processes. The metabolic epistemology proposed here seeks to provide such grounding.

The significance of this theoretical positioning lies not in replacing existing paradigms but in weaving them into an integrative architecture. Participatory, regenerative and intelligent design are reframed as partial articulations of a broader relational condition. Co-nature'ing consolidates these articulations within a single ontological and epistemological structure, preparing the ground for its operative translation in the spiral model.

The co-nature'ing spiral as operative framework

If Section Theoretical Foundations: Toward a Metabolic Epistemology establishes the epistemological foundations of Co-nature'ing, Section The Co-nature'ing Spiral as Operative Framework examines how this framework becomes operational within design practice. The Co-nature'ing spiral functions as the methodological translation of metabolic epistemology into a structured mode of engagement. Rather than prescribing a linear procedure, it offers a recursive architecture through which designers can navigate the distributed agencies that emerge across Human–Nature–Intelligence ecologies (Figure 2).

From linear processes to recursive ecologies

Most dominant design process models retain fundamentally linear or phase-based logics. The Double Diamond model structures divergence and convergence in alternating cycles of problem framing and solution development. Design Thinking frameworks typically move from empathy to ideation to prototyping and testing. Systems design approaches incorporate feedback loops but often remain anchored in human-centered problem definition.

These models have proven effective for complex problem-solving; however, they are built upon an implicit assumption: that human agents initiate and manage the process. Ecological and biodesign contexts challenge this premise. When living systems and biomaterials participate in shaping outcomes, design cannot be structured solely around human cognitive sequencing.

The spiral model diverges from phase-based approaches by foregrounding recursion rather than progression. Its structure reflects evolutionary dynamics: sensing, experimentation, systemic scaling and evaluative reflection continuously recalibrate one another. This recursive orientation aligns with complexity-informed learning and ecological cognition, where perception and action co-evolve.

Unlike linear iteration models, the spiral does not culminate in closure. Each cycle sustains and recalibrates perceptual engagement. In biodesign contexts – where materials grow, decay or transform – the assumption of definitive completion is untenable. The spiral therefore accommodates temporal openness as structural condition rather than anomaly.

The four modalities as relational orientations

The spiral is articulated through four interdependent modalities: Co-Sensing, Co-Growing, Co-Evolving and Co-Flourishing. These are not discrete stages but dominant orientations within an ongoing relational field.

To clarify their methodological role, each modality may be understood through a distinct operative emphasis: Co-Sensing concerns distributed ecological attunement; Co-Growing concerns adaptive material experimentation; Co-Evolving concerns systemic relational integration; and Co-Flourishing concerns relational regenerative evaluation. These emphases are interconnected but not interchangeable, as each modality foregrounds a different form of engagement within the spiral.

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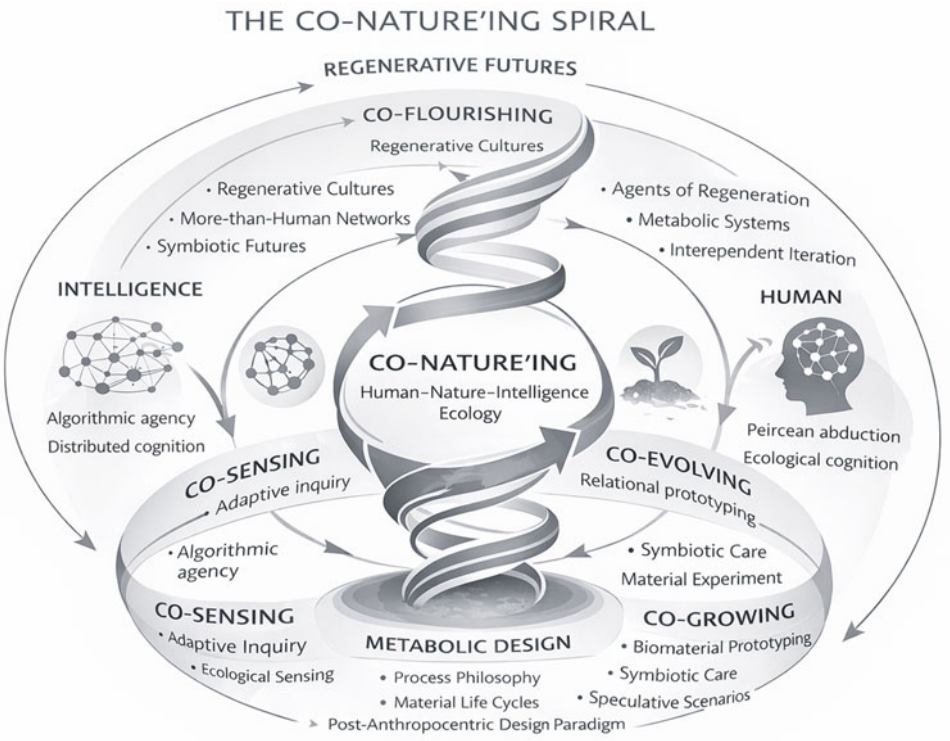


Figure 2. The Co-nature'ing spiral, developed by Américo Mateus and Susana Leonor. Source: The Authors

Co-sensing

Co-Sensing extends perception beyond human-centered inquiry. Traditional design processes begin with user research or stakeholder mapping. While these remain relevant, Co-Sensing expands perceptual engagement to include ecological signals, material affordances and data-driven insights as co-constitutive inputs.

In biodesign contexts, sensing may involve observing microbial growth patterns, monitoring humidity effects on biomaterials or analyzing algorithmically generated simulations. Perception becomes distributed across heterogeneous agents. As more-than-human design scholarship suggests (Giaccardi and Redström 2020; Nilsson et al. 2024), materials and organisms are not passive substrates but active participants in knowledge production.

Co-Sensing therefore functions as structured attentiveness. It resists premature solution framing and prioritizes relational observation. In pedagogical settings, this orientation recalibrates studio culture away from rapid ideation toward ecological literacy.

This orientation also resonates with emerging work on environmental sensing, multispecies attunement and more-than-human perception. Gabrys (2016) describes sensing infrastructures as active participants in environmental knowledge production, while Wakkary (2021) frames design as negotiation within more-than-human worlds. Myers (2015) similarly foregrounds embodied and ecological forms of attentiveness that exceed instrumental observation. Co-Sensing aligns with these perspectives while extending them toward the integration of biological and algorithmic mediation within biodesign practice.

Co-Sensing should therefore not be understood as computational crowd-sensing or distributed data acquisition alone, but as a relational condition of ecological attentiveness in which biological, material, human and algorithmic signals collectively shape perception.

Co-growing

Co-Growing translates relational perception into adaptive experimentation. In conventional prototyping, materials are shaped according to pre-established specifications. Within Co-Growing, experimentation unfolds as negotiation. Material behaviors, biological constraints and computational outputs inform iterative recalibration.

In biomaterial development, drying times, tensile resistance and microbial responses alter design trajectories. Prototyping becomes abductive: hypotheses are formed in response to emergent conditions rather than imposed a priori. Growth here denotes mutual adaptation rather than incremental refinement of a fixed concept.

Within this modality, speculative proposals also operate as relational mediators between biomaterial experimentation and future-oriented ecological imagination. By articulating "what if?" scenarios through biological materials, students are encouraged to critically project how emerging biofabrication practices could reshape everyday infrastructures, metabolic systems and relations between humans, nature and intelligence across possible futures. Co-Growing therefore links material experimentation with abductive speculation, shifting prototyping from formal optimization toward relational mediation.

This modality integrates principles from regenerative design and material-driven design but situates them within distributed cognition. Growth is not simply ecological metaphor; it is operational descriptor of reciprocal adjustment among interacting systems.

Co-evolving

Co-Evolving introduces systemic scaling and temporal projection. While Co-Growing focuses on localized experimentation, Co-Evolving situates projects within broader ecological and socio-technical infrastructures. Supply chains, energy flows, regulatory frameworks and territorial dynamics become integral to design consideration.

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Systems design traditions address such scaling; however, Co-Evolving distinguishes itself by maintaining recursive coupling with sensing and experimentation. Systemic mapping does not follow prototyping; it reshapes it. Design proposals are evaluated in light of regenerative capacity and infrastructural compatibility.

This modality is particularly relevant for biodesign, where material innovations intersect with bioeconomy strategies and environmental policy. Co-Evolving frames design not as isolated artifact production but as intervention within evolving ecosystems.

Co-flourishing

Co-Flourishing articulates the evaluative horizon of the spiral. Conventional assessment criteria emphasize usability, performance or market viability. Co-Flourishing reframes evaluation in terms of relational viability across domains.

Flourishing does not denote optimization but sustained systemic coherence. Biological vitality, social learning and technological mediation must reinforce rather than undermine one another. This evaluative orientation draws implicitly on ecological ethics while remaining open to contextual negotiation (Leopold 1949).

In educational contexts, Co-Flourishing expands critique sessions beyond esthetic or functional evaluation to include ecological integrity and long-term regenerative potential.

Flourishing is therefore not understood as permanent stabilization, but as the capacity of socio-ecological systems to sustain adaptive transformation, including material reintegration, decay and metabolic continuity across time.

AI and distributed intelligence within the spiral

The spiral explicitly accommodates algorithmic intelligence as a co-constitutive relational participant within design cognition. In AI-mediated design environments, computational systems generate patterns, simulate ecological scenarios and extend perceptual range (Bown 2021; Verganti et al. 2020). Rather than treating AI as an external tool, Co-nature'ing positions it within the Human–Nature–Intelligence triad. Predictive systems, generative AI models, simulation environments and algorithmic infrastructures increasingly participate in shaping ecological interpretation, speculative projection and biomaterial experimentation. Intelligence therefore operates not merely as technological augmentation but as an active epistemic condition within recursive ecological design processes.

However, this inclusion remains ethically and methodologically conditional. Distributed cognition does not imply symmetrical equivalence among Human, Nature and Intelligence. Computational mediation must remain embedded within ecological accountability. Agency within the framework is therefore distributed but asymmetrical: computational systems participate through predictive and generative capacities, biological systems through ecological and material transformation, while ethical accountability and critical mediation remain human responsibilities. By structuring AI engagement within recursive modalities rather than autonomous optimization, the spiral mitigates risks of technocratic determinism.

This positioning differentiates Co-nature'ing from AI-centric innovation models that foreground acceleration over relational integration.

Spiral as cognitive scaffold

The primary contribution of the spiral lies in its integrative function. It offers a cognitive scaffold that aligns practice with metabolic epistemology. Rather than isolating ecological awareness, participatory engagement and technological mediation as separate methodological streams, the spiral renders them structurally interdependent.

The recursive movement among sensing, experimentation, systemic reflection and relational evaluation creates a dynamic equilibrium. Each modality recalibrates the others. In this sense, the spiral functions less as method than as relational grammar.

For the biodesign community, this provides operational clarity. Living materials, biofabrication processes and hybrid intelligence systems demand process models capable of accommodating uncertainty, transformation and distributed agency. The Co-nature'ing spiral offers such a model without reducing complexity to procedural simplification.

A pedagogical instantiation: the BIP design roadmap

The theoretical and operative propositions of Co-nature'ing require empirical examination to assess their viability within educational practice. The BIP Design Roadmap (BIOFUTURES Week) functioned as a bounded experimental context in which the spiral framework structured a transdisciplinary biodesign studio. The program was conceived as an intensive, short-duration intervention designed to test whether metabolic epistemology could be enacted within pedagogical settings involving living materials and distributed intelligence systems.

Program structure and context

The BIP program brought together students from diverse design backgrounds, including communication design, product design and environmental studies. Participants engaged in a five-day intensive studio structured explicitly around the four modalities of the Co-nature'ing spiral. The program integrated field observation, biomaterial experimentation, systemic mapping and reflective evaluation (Van Ooijen-van der Linden and Griffioen 2023).

Algae were selected as the central material system due to their ecological significance, rapid growth cycles, material versatility and relevance to emerging bioeconomy strategies. Their dual status as living organism and material substrate positioned them as ideal catalysts for exploring the entanglement of biological and design processes. The pedagogical objective was not to produce finished products but to cultivate relational literacy across Human–Nature–Intelligence domains.

The studio environment included access to biomaterial preparation tools, environmental monitoring instruments and computational prototyping software. This configuration allowed biological experimentation and digital modeling to coexist within a single learning ecosystem (Naito & Botero 2024).

Methodological approach

The implementation of the spiral within the BIP program followed a structured yet adaptive sequence aligned with its four modalities. While not framed as a formal empirical study, the pedagogical intervention incorporated systematic observation, reflective documentation and qualitative synthesis to evaluate epistemic outcomes (Kolb 1984; Wang and Hannafin 2005).

Projects were evaluated according to their capacity to respect biological integrity, contribute to circular material flows, enhance ecological literacy and integrate technological mediation responsibly

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Figure 3. Co-Sensing activities during the BIP BIOFUTURES workshop, including ecological observation and algae material exploration-Day 1.
Source: The Authors



Figure 4. Co-Growing experimentation with algae-based biomaterials and adaptive prototyping processes during the BIP workshop – Day 2.
Source: The Authors

(Soares and Puccinelli 2023). Rather than functioning as discrete criteria, these dimensions operated as interdependent indicators of relational viability. Evaluation therefore moved beyond product performance toward systemic coherence, assessing whether proposals strengthened ecological cycles, supported distributed learning and maintained ethical accountability in the deployment of computational tools.

This qualitative synthesis focused not on product optimization but on shifts in cognitive orientation. The central research question guiding the pedagogical evaluation was whether students could move from solution-driven thinking toward relational mediation.

Operationalization of the spiral

Co-sensing in practice

The initial phase emphasized ecological and material observation prior to ideation. Students engaged in contextual mapping of algae's ecological roles, examining its functions in carbon sequestration, nutrient cycling and habitat formation. Material handling sessions allowed participants to directly observe texture, drying behavior, flexibility and structural limitations.

Rather than initiating immediate concept generation, students were required to articulate ecological interdependencies and material constraints. This reframed design inquiry from problem-

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solving to relational listening. Observational exercises were complemented by data analysis and computational visualization, introducing algorithmic mediation as perceptual extension rather than generative authority.

The result was a measurable delay in premature ideation, replaced by extended exploratory inquiry. In reflective sessions, participants reported heightened awareness of biological variability and environmental dependency.

For example, students initially approached algae-based materials through expectations of stability and predictability derived from conventional design practice. However, direct handling of the material revealed variations in texture, flexibility, drying behavior and structural response. These observations required students to slow down ideation and reformulate design decisions in response to material behavior, reinforcing Co-Sensing as a practice of ecological attentiveness rather than preliminary research (Figures 3 and 4).

Co-growing as iterative experimentation

The second modality centered on hands-on biomaterial prototyping. Students conducted controlled drying tests, binder experiments and small-scale structural trials. Material outcomes frequently diverged from initial expectations, requiring adaptive recalibration.

Rather than treating these deviations as failure, they were reframed as epistemic feedback. Prototyping became abductive: hypotheses emerged in response to material behavior. Iterative cycles revealed that growth-oriented experimentation fosters tolerance for uncertainty – a key competency in biodesign contexts where living systems introduce unpredictability.

One student team, for instance, initially attempted to stabilize algae-based composites through rigid formal control strategies derived from product design logic. Repeated deformation during drying cycles led the group to reformulate the project around adaptive flexibility rather than structural rigidity. In reflective discussion, this shift was described as moving from "controlling the material" toward "negotiating with the material," revealing a cognitive recalibration consistent with the relational premises of the framework.

Computational tools were selectively employed to simulate structural behavior or explore form variations. These tools functioned as relational participants within the spiral rather than as deterministic drivers.

Computational image-generation systems were also employed during speculative exploration phases. In one workshop activity, students used generative AI tools to visualize algae-based urban infrastructures and biofabrication scenarios, rapidly producing speculative iterations of possible ecological futures. Rather than accepting these outputs as authoritative proposals, participants critically evaluated the generated imagery against ecological feasibility, territorial conditions and material constraints. AI-generated visualizations therefore functioned as speculative mediators that expanded imaginative inquiry while remaining subject to ecological interpretation and relational accountability.

In addition to material experimentation, the Co-Growing modality incorporated a speculative design activity aimed at expanding the imaginative horizon of biomaterial prototyping. The exercise was adapted from the Speculative Proposals design resource developed by Palma and Neves (2025) and reformulated for a biofutures context. Students were invited to construct speculative scenarios exploring how biological materials – particularly algae – might reconfigure everyday practices, local infrastructures and relationships between humans, nature and technological systems. Within the studio, these scenarios functioned as narrative mediators linking biomaterial experimentation with broader imaginaries of metabolic futures.

This approach draws on speculative design traditions that employ "what if?" scenarios as critical devices for questioning dominant technological and social trajectories while opening space for alternative futures (Dunne and Raby 2013). Rather than seeking immediate solutions, the activity foregrounded the discursive and interrogative dimensions of design practice. Projects therefore operated closer to problematization than resolution. By articulating emerging material hypotheses with speculative projections of possible futures, the exercise reinforced the abductive and iterative character of Co-Growing. Prototyping was thus reframed not as formal optimization but as relational mediation, in which each experiment with biomaterials generated new speculative proposals while simultaneously recalibrating ecological, social and technological criteria of viability (Figures 5 and 6).

Co-evolving as systemic integration

In the third modality, projects were situated within broader ecological and infrastructural contexts. Students mapped supply chains, territorial implications and potential circular pathways for algae-derived materials. Discussions incorporated bioeconomy strategies, regulatory considerations and environmental impact.

This systemic orientation shifted project evaluation from object-level innovation toward ecosystem integration. Students were required to articulate how their proposals interacted with environmental cycles and socio-technical infrastructures.

The recursive nature of the spiral became evident at this stage: systemic mapping often prompted return to material experimentation, demonstrating the non-linear structure of the framework (Figure 7).

Co-flourishing as relational evaluation

The final modality reframed critique sessions around relational viability rather than isolated product performance. Projects were evaluated according to their capacity to respect biological integrity, contribute to circular material flows, enhance ecological literacy and integrate technological mediation responsibly. These dimensions were not treated as discrete checklist criteria but as interdependent indicators of systemic coherence. Evaluation therefore moved beyond esthetic refinement or functional optimization toward an assessment of how proposals participated within ecological cycles, supported distributed learning processes and maintained ethical accountability in the deployment of computational tools.

Flourishing, in this context, did not signify perfection or closure; rather, it denoted the degree to which human intention, biological vitality and algorithmic mediation could coexist without reinforcing extractive imbalance. This reframing encouraged students to consider long-term relational consequences, situating their projects within evolving socio-ecological infrastructures rather than treating them as isolated design artifacts (Figure 8).

Observed pedagogical effects

Three principal pedagogical outcomes emerged from the implementation.

First, the spiral structure moderated solution-oriented acceleration. Students demonstrated increased tolerance for ambiguity and delayed closure, consistent with relational epistemology.

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Figure 5. Iterative material testing and speculative biofutures development within the Co-Growing modality – Day 3.
Source: The Authors



Figure 6. Co-Evolving activities involving systemic mapping, biomaterial integration and ecological infrastructure reflection-Day 3 and 4.
Source: The Authors

Second, engagement with biomaterials fostered embodied understanding of ecological constraint. Rather than treating sustainability as abstract criterion, participants encountered biological processes directly, recalibrating their perception of material agency.

Third, the integration of computational tools within the relational framework prevented technological dominance. AI-supported simulations extended analytical capacity without displacing ecological attentiveness.

These outcomes were not understood as evidence of definitive competence acquisition, but as situated indicators of cognitive and pedagogical recalibration within the limits of a short intensive program. Their relevance lies in showing how students began to move from solution-oriented design habits toward more relational modes of observation, experimentation and evaluation.

The outcomes also suggests that the spiral model can structure studio environments in ways consistent with contemporary biodesign demands. While limited in temporal scale, the intervention demonstrates that metabolic epistemology can be enacted pedagogically without collapsing into procedural simplification.

Limitations and scope

The BIP program functioned as a short-duration intervention. Longitudinal studies would be required to assess sustained

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Figure 7. Co-Flourishing discussions and relational evaluation of biomaterial proposals within the BIP BIOFUTURES workshop – Day 5.
Source: The Authors

cognitive transformation or institutional integration. Additionally, the qualitative nature of evaluation limits claims of generalizability. The intent, however, was not statistical validation but epistemological demonstration.

Within this scope, the BIP Design Roadmap provides empirical grounding for the theoretical claims advanced in this paper. It demonstrates that relational ontology can be translated into structured educational practice, offering a viable pathway for embedding cognitive ecological frameworks within biodesign curricula.

Discussion: design as cognitive ecology in practice

The preceding sections articulated Co-nature'ing as both philosophical repositioning and operative framework and demonstrated its pedagogical instantiation through the BIP Design Roadmap. The discussion now turns to the broader implications of this framework for contemporary biodesign theory and practice. The central claim advanced in this paper is that design can be re-understood as a form of cognitive ecology – a distributed, relational mode of knowing and acting embedded within entangled biological, social and technological systems.

This claim is neither metaphorical nor rhetorical. It proposes a structural redefinition of design activity. If cognition is not confined to individual human minds but distributed across relational networks (Bateson 1972), and if living systems and computational infrastructures actively participate in shaping design outcomes, then design cannot remain framed as human-centered problem-solving. It must instead be conceptualized as mediation within dynamic ecologies of agency.

From human-centered design to distributed agency

Human-centered design has been foundational to contemporary practice. Its emphasis on empathy, participation and stakeholder engagement expanded design beyond object production of objects toward social responsibility. However, human-centered paradigms retain an implicit boundary: agency is redistributed among humans, while non-human entities remain contextualized as environment or resource.

More-than-human and posthuman design scholarship has challenged this limitation by foregrounding material and biological participation (Giaccardi and Redström 2020; Karana et al. 2023). Yet these approaches often operate as critical interventions rather than integrative epistemologies. They destabilize anthropocentrism but do not always provide structured operational alternatives.

Co-nature'ing extends this shift by proposing distributed agency as structural condition rather than theoretical critique. Within the Human–Nature–Intelligence triad, agency is distributed but asymmetrical (Latour 2005). This distinction is central to the framework. Biological systems, computational infrastructures and human actors participate through different modes of agency and cannot be treated as equivalent in intention, vulnerability or responsibility. Human intention, biological vitality and algorithmic mediation interact through continuous recalibration. The designer's role becomes one of mediator within these entanglements.

What distinguishes the Human–Nature–Intelligence triad from existing more-than-human and regenerative design frameworks is the explicit integration of computational mediation as an epistemic condition within design cognition (Nicenboim 2024). While more-than-human design scholarship productively expands agency toward biological and material systems, Co-nature'ing argues that predictive systems, generative AI models, simulation environments and data infrastructures increasingly participate in shaping ecological interpretation, speculative projection and material experimentation. Intelligence therefore operates not merely as digital augmentation but as a distinct relational participant within cognitive ecological processes.

This repositioning has consequences for authorship. Creative agency is no longer attributed solely to human cognition but

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Figure 8. BIP-Speculative and Performative Design for BIOFUTURES: Expanding Boundaries in Creative Practices and Concepts – Day 5. Source: The Authors

emerges through relational negotiation. Biological materials constrain and enable; computational systems generate alternatives and simulations; environmental conditions recalibrate feasibility. Design authorship becomes distributed and situational.

Such a shift does not diminish human responsibility. On the contrary, it intensifies it. If design outcomes are co-produced by multiple agencies, designers must account for the relational consequences of their mediation. Ethical evaluation expands accordingly.

Evaluation beyond performance metrics

In conventional design discourse, evaluation frequently centers on usability, esthetic coherence, market viability or environmental efficiency. Regenerative design broadened this scope by introducing restoration and resilience as criteria (Hecker and Toivonen 2024). However, these metrics often remain project-specific or outcome-focused.

The Co-nature'ing framework introduces relational viability as a complementary evaluative orientation. Flourishing, as articulated

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within the spiral, is not synonymous with optimization. It refers to configurations in which interactions across Human–Nature–Intelligence domains reinforce systemic coherence rather than generate extractive imbalance.

This evaluative shift reframes sustainability from compliance metric to relational condition. Rather than asking whether a design reduces environmental impact, the question becomes whether it participates constructively within ecological cycles and socio-technical infrastructures. Similarly, rather than evaluating AI integration solely for efficiency gains, one must consider whether computational mediation enhances or undermines ecological awareness.

By positioning evaluation within distributed cognition, Co-nature'ing encourages designers to move beyond isolated performance indicators toward systemic integrity.

Temporality and recursion

A further implication concerns temporality. Industrial production models often assume linear progression: problem identification, solution development, deployment. Even iterative design models typically assume convergence toward stabilization.

Biodesign disrupts this temporality. Living materials grow, decay and transform. Ecological systems evolve beyond project timeframes. AI systems update continuously. Under such conditions, closure becomes provisional.

The spiral's recursive structure accommodates this instability. Sensing, experimentation, systemic reflection and evaluation are not sequential phases but ongoing recalibrations. Each modality may prompt return to earlier engagements. In practice, this fosters tolerance for uncertainty and adaptive responsiveness – competencies essential for working with living systems.

Understanding design as cognitive ecology therefore implies recognizing temporality as evolutionary rather than project-bound. Design becomes ongoing participation within changing systems rather than discrete intervention.

Within this recursive temporality, design increasingly operates as a condition of co-habitation rather than episodic intervention. Human, biological and computational agencies do not merely collaborate toward isolated outcomes but continuously negotiate shared ecological, material and cognitive environments across time. Co-habitation therefore foregrounds the durational dimension of Co-nature'ing, emphasizing sustained coexistence, mutual adaptation and relational maintenance within evolving socio-ecological systems.

Within biodesign contexts, temporality also implies acknowledging decomposition as an active phase of material existence rather than as project termination. Living materials continue to transform beyond the temporal boundaries of conventional design cycles, suggesting that decay itself can operate as a designed and anticipated dimension of relational life. Designing for decomposition therefore extends the recursive logic of the spiral into biological afterlives, framing material dissolution as metabolic reintegration within ecological systems rather than as waste or failure.

AI and the risk of technocratic drift

The integration of algorithmic intelligence into design practice introduces both opportunity and risk. Generative AI systems extend perceptual range, simulate ecological scenarios and accelerate iteration (Bown 2021; Verganti et al. 2020). However, uncritical adoption may reintroduce technocratic dominance under the guise of distributed agency.

Technocratic drift is particularly significant within biodesign contexts because computational optimization can easily privilege efficiency, scalability or predictive control over ecological responsiveness and situated material knowledge. AI-assisted systems may unintentionally suppress biological variability by favoring standardization and calculability. Co-nature'ing therefore proposes reflexive mediation practices in which computational outputs remain continuously interpreted through ecological observation, material feedback and collective critical reflection.

If computational outputs are privileged as objective optimization, ecological and ethical dimensions may be subordinated. The Co-nature'ing framework addresses this risk by embedding AI within relational accountability. Algorithmic mediation is positioned as participant rather than authority. Its outputs remain subject to ecological and social interpretation.

This positioning is particularly relevant for biodesign, where AI-assisted biomaterial simulation and environmental modeling are increasingly prevalent. Without epistemological grounding, such tools risk reinforcing extractive acceleration. Within cognitive ecology, they become instruments of expanded relational awareness.

The recognition of distributed agency does not dissolve ethical responsibility into generalized systems interaction. On the contrary, it intensifies the designer's accountability as mediator among heterogeneous agencies operating under unequal capacities and vulnerabilities. Co-nature'ing therefore positions ethical responsibility not as centralized authorship but as relational stewardship across evolving ecological and technological systems.

Institutional and pedagogical implications

The BIP instantiation suggests that cognitive ecological structuring can be introduced within educational contexts. However, scaling such frameworks raises institutional questions. Academic structures often reward rapid output, disciplinary specialization and product-oriented assessment. Recursive relational models require extended reflection, interdisciplinary integration and tolerance for ambiguity.

The recursive temporality proposed by Co-nature'ing also exposes a structural tension with conventional academic calendars. Semester-based studio models privilege fixed deliverables, compressed timelines and accelerated production cycles, whereas living materials and ecological processes often evolve according to slower and non-linear temporalities. Future biodesign pedagogies may therefore require longer experimental cycles, open-ended laboratory engagement and extended ecological observation periods capable of accommodating the temporal conditions of living systems. The recursive temporality proposed by Co-nature'ing also exposes a structural tension with conventional academic calendars. Semester-based studio models privilege fixed deliverables, compressed timelines and accelerated production cycles, whereas living materials and ecological processes often evolve according to slower and non-linear temporalities. Future biodesign pedagogies may therefore require longer experimental cycles, open-ended laboratory engagement and extended ecological observation periods capable of accommodating the temporal conditions of living systems.

Embedding Co-nature'ing within curricula would therefore necessitate structural adaptation: revised assessment criteria, integration of biomaterial laboratories and explicit engagement with distributed intelligence systems. Governance frameworks must also align with relational evaluation metrics.

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While the present study does not provide longitudinal institutional data, it demonstrates that metabolic epistemology can structure studio practice without collapsing into procedural formalism. This provides a foundation for future research on curricular integration and institutional transformation.

Beyond its immediate pedagogical function, the speculative activity revealed an important dimension of the Co-nature'ing framework: its capacity to cultivate future-oriented reflexivity in design practice. By inviting students to construct speculative biofutures through "what if?" scenarios, the exercise foregrounded how present design decisions participate in shaping both desirable and undesirable futures (Dunne and Raby 2013). Rather than approaching biomaterial experimentation purely as a technical or esthetic problem, students were encouraged to reflect on the broader socio-ecological implications of their projects. This perspective resonates with accounts of design as a future-making and socially transformative praxis (Light 2015; Malpass 2017), positioning designers not as neutral problem-solvers but as agents implicated in eco-social change. Situating everyday practices and material choices within extended temporal horizons encouraged reflection on the consequences of action and inaction, while activating the capacity to think with what does not yet exist. In this sense, speculative biofutures functioned as cognitive devices that supported the abductive and exploratory logic of the Co-nature'ing spiral. Moreover, the activity aligns with transition design perspectives that frame design as a catalyst for long-term societal transformations, connecting local everyday practices with broader multi-temporal transitions toward more regenerative futures (Irwin 2015).

Limitations and future directions

Several limitations warrant acknowledgment. First, the pedagogical instantiation was short in duration, limiting capacity to assess long-term cognitive transformation. Second, qualitative evaluation methods provide interpretive insight but not statistical generalizability. Third, the framework remains theoretical in its institutional implications; large-scale implementation requires further empirical investigation.

Future research should explore longitudinal integration of spiral-based curricula, industrial applications within biofabrication contexts and governance models capable of embedding relational evaluation at policy level. Comparative studies with regenerative and posthuman design frameworks would further clarify the distinct contribution of metabolic epistemology.

Despite these limitations, the conceptual contribution is significant. By integrating participatory, regenerative and AI-mediated design discourses within a unified relational ontology, Co-nature'ing advances biodesign beyond thematic alignment toward epistemological coherence.

Conclusion

This paper has argued that biodesign operates within epistemic conditions that exceed anthropocentric and instrumental paradigms. As living materials, ecological systems and algorithmic infrastructures increasingly participate in shaping design processes, the discipline must confront the limitations of linear, human-centered frameworks. The conceptual challenge is not simply to incorporate new actors into existing models, but to reconceptualize design itself as a relational practice embedded within distributed systems of agency.

Within this condition, agency cannot be understood as either singular or equivalent. Human, biological and computational systems participate through asymmetrical forms of mediation, transformation and responsibility that continuously reshape the conditions of design practice.

Co-nature'ing has been proposed as a response to this challenge. Grounded in process philosophy, complexity theory, ecological cognition and contemporary more-than-human design research, it articulates a metabolic epistemology in which Human, Nature and Intelligence are understood as interdependent domains. Rather than positioning biodesign as a thematic extension of sustainability discourse, the framework redefines it as cognitive ecology: a practice of mediated participation within entangled biological, social and technological systems.

The spiral model translates this epistemology into operative structure. Through recursive modalities of Co-Sensing, Co-Growing, Co-Evolving and Co-Flourishing, it provides a relational scaffold capable of accommodating living materials, systemic scaling and AI-mediated experimentation. The pedagogical instantiation through the BIP Design Roadmap demonstrates that such a framework can be enacted within educational environments while preserving the relational complexity inherent to living systems. While limited in scope and duration, this instantiation provides empirical grounding for the viability of metabolic epistemology in practice.

The contribution of this paper lies not in proposing a closed methodology, but in articulating a relational epistemological framework capable of connecting ecological cognition, living materials and computational mediation within contemporary biodesign practice. Participatory, regenerative and AI-mediated design discourses are not treated as competing paradigms but as partial articulations of a broader relational condition. Co-nature'ing synthesizes these trajectories within a shared relational epistemological framework, offering conceptual coherence for interdisciplinary research involving living systems and hybrid intelligence.

Future work must investigate how cognitive ecological frameworks can scale beyond pedagogical interventions into institutional, industrial and governance contexts. As biodesign continues to expand across domains, the need for epistemological clarity will intensify. Co-nature'ing offers one pathway toward such clarity, reframing design not as mastery over systems, but as structured participation within them.

In this sense, biodesign increasingly becomes less a practice of producing stable solutions and more a recursive practice of cohabitation within evolving ecological, material and intelligent systems.

Data availability statement. The data supporting the findings of this study consist primarily of pedagogical documentation, workshop materials, project outputs and observational notes generated during the implementation of the Blended Intensive Programme (BIP) described in the paper. These materials were produced within the context of educational activities conducted at ISMAT and collaborating partner institutions. Due to their nature as teaching materials and student-generated work, the data are not publicly archived but may be made available by the authors upon reasonable request.

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